

Strength in Numbers: Interstate Purchasing Pools and Medicaid Supplemental Rebate Capture

Jonathan Palisoc

University of Michigan

July 2026

Abstract

Objective: To examine whether joining an interstate Medicaid drug purchasing pool raises the supplemental rebates states capture from drug manufacturers; specifically, whether joining the Sovereign States Drug Consortium (SSDC), the largest purchasing pool owned and governed by its member states, raises the supplemental rebate share of drug spending. Supplemental rebates are voluntary discounts negotiated above the federal statutory floor, and there is almost no causal evidence on whether negotiating as a bloc raises them.

Data Sources: CMS-64 supplemental rebate data linked to a primary-source panel of pool membership for all 50 states, 2002–2024, reconciled from state Medicaid manuals, archived web materials, agency correspondence, and public-records requests.

Study Design: Staggered difference-in-differences, event study models; the eight states that adopted SSDC in 2012 or later compared to the thirteen states that never joined any pool; estimators built for staggered adoption; wild cluster bootstrap inference given the small number of treated states.

Principal Findings: Later adopters saw large gains in rebate capture after joining SSDC. States that later joined SSDC were capturing *less* than their never-pooled peers before joining (1.33 versus 2.31 percent of drug spending), and their capture rises steadily after entry. The difference-in-differences estimates confirm the visible catch-up: joining SSDC raised the supplemental

rebate share of drug spending by 3.4 percentage points among the eight later adopters (wild cluster bootstrap $p = 0.051$). The estimate is stable across five estimators and leave-one-state-out checks, and it survives a state-by-state check of co-timed pharmacy-benefit reforms. Smaller states made the largest gains, about twice those of larger states, consistent with a countervailing-buyer-power mechanism. The gains arrived primarily through the SSDC: the two vendor-administered pools show smaller effects or effects that cannot be distinguished from zero, and pooling of any kind averages one to two points of drug spending. Precision is limited for a structural reason (by 2024 only thirteen states had never pooled), and the outcome is rebate *capture*, not net drug spending, so the gains should not be booked as guaranteed savings. The findings still carry immediate policy weight: with the 2025 federal reconciliation law withdrawing roughly \$911 billion in federal Medicaid financing over the coming decade, most cost-containment levers available to states cut eligibility, benefits, or provider payment, while pooled rebate negotiation extracts resources from manufacturers instead. The 50-state membership panel is released as a public good.

1. Introduction

1.1 Rising Medicaid Drug Costs and the Search for Fiscal Levers

Medicaid is the single largest payer for prescription drugs in the United States. In fiscal year 2023, gross Medicaid drug spending was over \$80 billion, with net spending after rebates reaching roughly \$51 billion and continuing to rise as specialty and biologic therapies absorbed a growing share of state budgets (MACPAC, 2022). Net Medicaid drug spending increased 72 percent between FY2017 and FY2023, outpacing overall Medicaid spending growth and placing

escalating pressure on both state and federal budgets. As such, managing pharmaceutical expenditures is a first-order fiscal challenge for state Medicaid programs.

The structure of modern pharmaceutical markets makes that challenge harder. Specialty and biologic drugs account for around 2 percent of prescriptions but close to 50 percent of drug spending by the early 2020s. Unlike traditional small-molecule drugs, which eventually face generic competition that drives down prices, many biologics face limited biosimilar entry and sustained pricing power. The emergence of cell and gene therapies with per-patient costs exceeding \$1 million has introduced entirely new categories of fiscal risk. State Medicaid agencies must simultaneously contain growth in drug spending, maintain compliance with federal coverage rules, and preserve access for populations that rely heavily on public insurance. The fiscal challenge has now sharpened abruptly. The 2025 federal budget reconciliation law (H.R. 1), enacted in July 2025, is projected to reduce federal Medicaid spending by roughly \$91 billion over the coming decade, or about 14 percent of projected federal Medicaid outlays, and the Congressional Budget Office expects the law's health provisions to increase the number of people without insurance by roughly ten million (Euhus, Williams, Burns, and Rudowitz, 2025; KFF, 2025). States are already bracing. In the first annual Medicaid budget survey after enactment, states reported mounting budget pressure, an uptick in provider-rate restrictions, and near-universal new or expanded initiatives to contain prescription drug costs (Hinton et al., 2025). Most of the levers available to a state absorbing lower federal financing (tightening eligibility, trimming benefits, cutting provider payment) conserve dollars by reducing coverage or access. Extracting larger rebates from manufacturers is one of the few that does not. Whether banding together with other states actually strengthens that lever is therefore no longer a question

about a niche financing stream - it bears directly on how states absorb a major federal Medicaid retrenchment while preserving access.

State Medicaid programs operate within a mixed federal-state structure that shapes their cost-containment options (Section 2.1). At the state level, agencies retain meaningful discretion over several levers: preferred drug list (PDL) design, prior authorization rules, managed-care pharmacy architecture, drug utilization review, and supplemental rebate negotiations. This paper focuses on the last of those levers, and on one increasingly common attempt to strengthen it: banding together with other states to negotiate as a bloc.

1.2 Supplemental Rebates as an Understudied Fiscal Lever

Supplemental rebates are voluntary rebates negotiated above and beyond the federal Medicaid Drug Rebate Program (MDRP) minimum. Manufacturers pay them in exchange for preferred formulary placement, easier utilization management, or broader access to a state's covered population. As of 2019, 47 states and the District of Columbia had supplemental rebate agreements, which collectively generated over \$1.7 billion annually (OIG, 2014; Gifford et al., 2020).

Despite that fiscal importance, supplemental rebates have received very little rigorous empirical attention. The Office of Inspector General documented substantial cross-state variation in how states collect and structure supplemental rebates but did not attempt to explain it (OIG, 2014). MACPAC (2018) estimated that supplemental rebates account for roughly 6 percent of total Medicaid drug rebates but did not investigate the drivers of variation. The academic Medicaid pharmacy literature has focused on adjacent policies (managed care versus fee-for-service drug spending, formulary design, prior authorization, generic substitution), leaving supplemental rebates as one of the largest policy-relevant blind spots in the evidence base.

That omission matters because the cross-state variation is large. In 2024, supplemental rebates ranged from near zero in some states to over 28 percent of drug spending in others. If that variation reflects replicable institutional arrangements, the policy implications are substantial. If it instead reflects idiosyncratic state characteristics or bundled reforms that cannot be transferred, the lesson is very different. Distinguishing between these explanations requires causal evidence, which to date does not exist.

1.3 Interstate Drug Purchasing Pools

Many states attempt to increase bargaining leverage by joining interstate drug purchasing pools. The intuitive logic is straightforward: a larger covered population should strengthen the threat to steer utilization toward preferred products, which should in turn raise supplemental rebates.

Three major pool families emerged in the mid-2000s: the National Medicaid Pooling Initiative (NMPI, established 2003), The Optimal PDL Solution (TOP\$, established 2005), and the Sovereign States Drug Consortium (SSDC, established 2005). By 2024, 33 of 50 states participated in one of those three arrangements.

The theoretical case for pooling is plausible. Buyer-side market power models suggest that larger, more coordinated purchasers can secure better prices when they can credibly redirect demand (Chown, Dranove, Garthwaite, and Keener, 2019). Structural models of formulary-rebate contracting imply that rebates depend on covered lives and on the buyer's ability to commit to preferred placement (Ho and Lee, 2024). International pooled-procurement research reaches a similar conclusion: pooling can improve procurement outcomes, but the size of the gain depends heavily on governance and implementation (Dubois, Lefouili, and Straub, 2021; Parmaksiz et al., 2022).

However, despite those strong priors, there is no settled causal evidence on whether interstate Medicaid purchasing pools actually increase supplemental rebate capture. Existing accounts are entirely descriptive. Horvath (2019) documented that Vermont reported 4.7 percent additional savings in its first year of SSDC participation and that New York reported \$80.5 million in additional savings through NMPI, but these figures lack counterfactual comparisons. In short, two-thirds of U.S. states have adopted a policy intervention for which the published evidence base consists almost entirely of anecdotes and self-reported savings figures.

This paper puts the state-owned SSDC at the center of the analysis. Among the three arrangements, SSDC has the strongest institutional claim to converting pooled covered lives into bargaining power (its member states own the contracts, govern the pool, and decide what to accept), and its staggered adoption by fifteen states through 2024 provides the cleanest quasi-experimental variation. The two vendor-administered pools serve as secondary comparisons. Figure 1 sketches the pooling mechanism: scale raises capture only when paired with credible volume steering (Section 2.4).

1.4 Contribution

This paper adds to the literature in a few ways.

First, it constructs a 50-state primary-source panel of Medicaid drug-pool membership from 2002 through 2024. No pre-existing public dataset provides a fully reconciled state-year history of SSDC, NMPI, TOP\$, and no-pool participation, and the secondary compilations that circulate disagree on the membership and timing of many states. I reconcile the record against primary sources (current and archived state Medicaid manuals, agency correspondence, CMS supplemental-rebate-agreement tables, and public-records requests) and document each coding decision. The panel is released as a public good.

Second, it provides the first quasi-experimental evidence on whether joining SSDC raises supplemental rebate capture, with parallel estimates for the vendor-administered pools as secondary comparisons. The estimators are matched to the institutional facts, and the inference is built for the small number of treated states rather than borrowed from large-sample defaults (Section 4.3; estimator and inference mechanics are collected in Appendix G).

Lastly, it maps a countervailing-buyer-power mechanism to testable heterogeneity by pool type and state size, and it bounds the estimates honestly given the small number of never-pooled comparison states. Joining SSDC raises the supplemental rebate share of drug spending by about 3.4 percentage points among later adopters. Averaged across all pools the gain is a more modest one to two points, and the vendor-administered pools on their own show smaller or statistically indistinguishable effects. The pattern is consistent with a countervailing-buyer-power mechanism (Section 2.4): a modest positive effect, though imprecisely estimated.

The core identification problem here is not simply that some states are permanently different from others. It has two parts. First, states often change multiple pharmacy-benefit features (adopting a uniform PDL, carving pharmacy out of managed care, changing who negotiates rebates) at roughly the same time they join a pool. Second, and more basic, “joining a pool” must be measured accurately before any control strategy can help. This paper addresses the first problem with a transparent, symmetric control policy (Section 4.2) and the second by measuring pool membership directly from primary sources (Section 3).

The remainder of the paper proceeds as follows. Section 2 reviews institutional background and develops the bargaining theory, Section 3 describes the data, and Section 4 presents the empirical strategy. Section 5 reports results, Section 6 discusses implications and limitations, and Section 7 concludes.

2. Background and Theory

2.1 The Medicaid Drug Rebate Program

A short primer on Medicaid pharmacy benefits is useful before the statute. Each state's Medicaid pharmacy benefit operates within a layered federal-state architecture that determines how drug spending and rebates flow. At the federal level, the MDRP sets a statutory rebate floor: manufacturers must rebate a minimum share of every Medicaid prescription in exchange for Medicaid coverage of their products. At the state level, state Medicaid agencies decide which drugs to make easier or harder for clinicians to prescribe, primarily through the Preferred Drug List. The PDL is a state-published list of which drugs are "preferred" (no prior authorization, easier prescribing) and which are non-preferred (prior authorization required, sometimes a step-therapy fail-first requirement). Manufacturers compete for preferred placement by offering supplemental rebates above the federal MDRP minimum. Preferred placement materially increases a drug's Medicaid market share within its therapeutic class, so manufacturers are willing to pay for it. The state's bargaining leverage in supplemental rebate negotiations is, mechanically, a function of how many covered lives it can credibly steer with a PDL decision. Two further architectural choices shape that leverage. Pharmacy benefit carve-in versus carve-out determines whether the pharmacy benefit is administered by the state directly (carve-out, where the state controls the PDL and rebates for the entire pharmacy benefit, including for managed-care enrollees) or by managed-care organizations on contract with the state (carve-in, where MCOs run their own formularies for their enrolled members, with statewide coordination varying widely). Uniform PDL policies require all MCOs to use the state's PDL rather than negotiate their own, converting a fragmented set of MCO formularies back into a single

statewide formulary decision. A state can sit in any of the four combinations of uniform PDL and carve-out status, and the size of its credible bargaining unit varies accordingly. Interstate purchasing pools, the policy lever this paper studies, sit on top of all of this: they combine multiple states' covered lives into a single negotiating unit. Whether pool membership translates into higher supplemental rebate capture should depend on the state's ability to translate pooled covered lives into a coordinated PDL decision, which in turn depends on the carve-out and uniform-PDL architecture.

The MDRP itself was established by the Omnibus Budget Reconciliation Act of 1990 and has been a central feature of Medicaid pharmaceutical policy for more than three decades.

Manufacturers enter rebate agreements with the Secretary of Health and Human Services; for brand-name drugs, the federal unit rebate amount is the greater of 23.1 percent of Average Manufacturer Price (AMP) or the difference between AMP and the manufacturer's best price (the lowest price offered to any commercial purchaser, with certain exclusions), plus an inflation penalty when price growth exceeds CPI-U. For generics, the statutory minimum is 13 percent of AMP.

This pricing structure creates well-documented incentive distortions: Duggan and Scott Morton (2006) showed that manufacturers respond to the best-price provision by strategically raising prices to commercial purchasers, so the MDRP's mandatory rebates are partially offset by higher commercial prices.

The best-price provision has an additional consequence for supplemental rebate negotiations. Because manufacturers must offer Medicaid a rebate at least as large as the AMP-best-price spread, any price concessions to large commercial purchasers automatically increase the statutory rebate. This creates a floor below which supplemental negotiations operate: the

supplemental rebate is an increment above an already-deep statutory discount, which makes the marginal dollar obtained through supplemental negotiation harder to extract. The implication for this paper is that a few-percentage-point supplemental gain is economically meaningful, not trivial, because it is extracted on top of a binding federal floor.

The Affordable Care Act intensified the MDRP's role in 2010 through two changes relevant to this analysis. First, it raised the minimum brand-name rebate from 15.1 to 23.1 percent of AMP, substantially raising the statutory floor. Second, it extended mandatory MDRP rebates to drugs dispensed through Medicaid managed care, closing a loophole that had allowed MCOs to negotiate outside the MDRP framework. After 2010, supplemental rebates could be layered on top of statutory rebates for both fee-for-service and MCO-dispensed drugs. This increased the potential scope of supplemental rebate programs but also added administrative complexity.

2.2 Supplemental Rebates and Preferred Drug Lists

Supplemental rebates are embedded in formulary design, prior authorization policy, and the state's ability to steer utilization toward preferred products.

The economics of supplemental rebates are best understood through formulary contracting theory. Ho and Lee (2024) developed a structural model of formulary-rebate contracting between PBMs and manufacturers, showing that multi-tier formularies substantially increase negotiated rebate payments. In their model, rebate contracts specify per-unit payments contingent on formulary placement and competitor positioning. A manufacturer whose drug is placed on the preferred tier gains market share within its therapeutic class, while the payer captures a rebate that partly offsets coverage costs. The size of the rebate depends on the manufacturer's outside option (losing preferred status) and the payer's outside option (switching to an alternative). That logic implies that supplemental rebate negotiation is strongest in therapeutic classes with credible

alternatives and weakest in sole-source or minimally substitutable products - a class-level prediction I return to and test, coarsely, in Section 5.6.

Chown, Dranove, Garthwaite, and Keener (2019) frame the same intuition in their analysis of buyer-side market power in health care. Comparing the United States and Canada, they argue that prescription drug price differences more directly reflect public buyer power than provider-wage differences do.

Preferred drug lists are the operational mechanism through which that leverage is exercised.

Virabhak and Shinogle (2005) documented large prescribing responses to PDL implementation: restricted-drug prescription shares fell by 9.0 percentage points in Illinois Medicaid and 6.2 percentage points in Louisiana Medicaid after PDL adoption. Munshi et al. (2018) found that Florida's state-mandated PDL reduced overall drug use by 6 percent but increased plan costs by 27 percent. This apparently counterintuitive result reflects the PDL channeling utilization toward preferred branded drugs with high acquisition costs but also high rebate yields. It illustrates that PDL effects on spending are not straightforward: a PDL that maximizes supplemental rebate revenue may not minimize net drug costs if it channels utilization away from lower-cost generics. It also clarifies why this paper's outcome is rebate *capture* and not net spending. A state with a uniform PDL (Section 2.1) can credibly commit to placement decisions for a larger share of its covered lives, which should increase bargaining power.

2.3 Interstate Drug Purchasing Pools

Three interstate pool families operate during the panel: SSDC, NMPI, and TOP\$ (Section 1.3).

The distinction that matters for this paper is who owns the bargaining. SSDC is, in its own words, "the only Medicaid rebate pool organized and managed by the states." Member states contract *individually* with manufacturers using their own state-specific supplemental rebate

agreements while negotiating with the leverage of the consortium's combined covered lives, retain full ownership of their contracts and preferred drug lists, and make final acceptance decisions through an annual member-state cycle. NMPI and TOP\$ are instead vendor-administered: a commercial administrator negotiates collective rebate agreements on the member states' behalf and owns the contract paper, with participation running through the state's services agreement with the vendor. By 2024, 15 states participated in SSDC, 11 in NMPI, and 7 in TOP\$.

This structural contrast, state-owned pooling with individual contracts versus collective contracting administered by a vendor, is the basis for the pool-type comparison this paper makes. The next subsection develops why it should matter; further institutional detail on the vendor pools is collected in Appendix B.

2.4 Theory: Pooling, Countervailing Power, and Steerable Volume

The economic logic of pooling is simple. What a manufacturer will pay for preferred formulary placement depends on what it stands to lose without it: the volume a formulary decision can credibly move away from its drug, times the margin on that volume. A state bargaining alone puts only its own steerable volume behind the threat of non-preferred placement. A pool puts the combined bloc's volume behind it, which worsens the manufacturer's no-deal outcome and raises the rebate the states can extract. (Appendix K states this argument as a compact Nash bargaining model.) Three implications follow directly and structure the empirical work.

First, size alone is not enough. The threat is only as good as the volume the pool can actually steer. A pool that assembles nominal enrollment but cannot coordinate formulary action, because PDL decisions are fragmented across managed-care plans or the pharmacy benefit is carved into managed care with autonomous formularies, adds covered lives without adding leverage. Pooling

and *steerable* volume are complements. This is why governance (uniform PDLs, carve-outs, MCO rebate collection) is part of the mechanism, not a confounder to be adjusted away indiscriminately. Figure 2 sketches these governance levers - the within-state arrangements that concentrate formulary leverage into a single, credible volume signal that a pool can put behind its negotiation.

Second, smaller states should gain more. A small state bargaining alone has little volume to threaten with; pooling relaxes that constraint most where it binds hardest. The model therefore predicts a larger percentage-point gain for smaller states, which is a testable heterogeneity prediction.

Lastly, pool governance should matter. SSDC is state-owned: the parties who keep the rebate savings also control the bargaining (Section 2.3), and its state-specific agreements let manufacturers offer deeper concessions to states that can credibly move a therapeutic class without extending the same terms to states that cannot. Vendor-administered pools rely on a contractor whose objectives (client retention, standardized operations, manufacturer relationships) need not rise and fall with each incremental rebate dollar. This is an incentive-alignment hypothesis, not an accusation of shirking, and it predicts that SSDC should at least match the vendor pools. A related class-level prediction, that pooling should help most where close therapeutic substitutes make the steering threat valuable, is tested coarsely in Section 5.6. Two cautions temper the theory. First, the same features that should make SSDC effective also shape who selects into it. SSDC membership skews toward states with unified FFS-MCO PDLs and MCO rebate collection, and pool entry often travels with exactly those reforms. Part of a measured “SSDC effect” may therefore reflect the correlated bundle rather than pool governance in isolation. I read the estimates as the effect of SSDC *as implemented* (Section 4.1), and the

design confronts the bundling directly (Section 5.4) rather than assuming it away. Second, a rising measured rebate share can reflect either genuinely stronger bargaining or better administrative capture and reporting; the co-timed reform check in Section 5.4 exists largely to address this hazard (see also Section 6.3). Appendix K lays out the full causal chain, including which variables are pre-treatment confounders and which are post-treatment mediators, underlying the control policy in Section 4.

2.5 International and Pharmacy-Benefit Evidence

International pooled-procurement evidence reinforces the steerable-volume logic. Dubois, Lefouili, and Straub (2021) study centralized procurement across seven low- and middle-income countries and find that pooled public procurement is associated with lower drug prices, with smaller reductions when the supply side is more concentrated. Parmaksiz et al. (2022), in a systematic review of 44 empirical studies, similarly emphasize the capacities and alignment needed among buyers, pooled-procurement organizations, and suppliers. Savings range widely across settings, which suggests that pool design features, not pooling per se, drive outcomes. The lesson maps onto Medicaid: scale matters less when the buyer cannot credibly move volume. The interaction between pharmacy-benefit design and supplemental rebate capture is also well grounded. Dranove, Ody, and Starc (2021) provide the most rigorous causal study of Medicaid pharmacy-benefit privatization, finding that full privatization decreased drug spending by 22.4 percent, with effects concentrated in states giving MCOs formulary flexibility. This implies that MCO formulary autonomy can either complement or undermine pool-negotiated PDLs. Bendicksen and Kesselheim (2022) document a countervailing trend of pharmacy-benefit carve-outs. Hernandez and Gellad (2020) show that fee-for-service programs tend to retain expensive branded drugs to capture large inflation rebates, while MCOs adopt generics at much higher rates

- relevant here because pool-negotiated PDLs may favor branded drugs with large supplemental rebates over lower-cost generics. These reforms move with pool entry and are the reason a naive pool coefficient must be interpreted carefully. However, as Section 4.2 argues, they are as much mechanism as confounder.

***## 3. Data

3.1 CMS-64 Financial Management Reports

The primary outcome is annual supplemental rebate capture, defined as supplemental rebate dollars divided by total drug expenditures. Both the numerator and the denominator come from CMS-64 Financial Management Reports. I use a share rather than a dollar outcome because states differ enormously in program size; the share-based outcome asks whether a state is extracting larger supplemental rebates relative to the scale of its own drug program.

Constructing a consistent CMS-64 panel requires harmonization across multiple reporting formats and line-item conventions. The raw data files span multiple formats: one master file covers fiscal years 2002–2011, while individual annual files cover later years. I compiled all years into a master file with standardized variable names and consistent state identifiers. A total of 39 state-year observations (3.4 percent of the panel) have missing supplemental rebate data, which I leave as missing rather than impute. Six state-years have negative supplemental rebate values, which is conceptually difficult to reconcile with voluntary rebate payments. Four are negligible in magnitude, while North Carolina in 2013 (–8.1 percent) and Virginia in 2012 (–3.0 percent) are larger and likely reflect timing or reporting artifacts. These observations are retained without adjustment; excluding them does not materially affect the results.

After standardization, the rebate panel contains 1,150 state-years from 2002 through 2024, with 1,111 nonmissing supplemental rebate observations. The CMS-64 reports on a fiscal-year basis,

which may not align perfectly with calendar-year pool membership indicators; Section 6.3 discusses this misalignment.

3.2 State Drug Utilization Data

State Drug Utilization Data (SDUD) enter the analysis in two ways. First, they serve as an external validation source for the CMS-64 outcome. Total drug spending figures are cross-referenced against the CMS-64 to identify states where the two sources diverge substantially. Some discrepancies are expected given the two sources' different reporting scope and timing. Second, SDUD provides state-year drug-mix measures used in outcome-construction robustness checks and in the class-substitutability heterogeneity test. The SDUD mix panel summarizes each state-year into several descriptive measures: single-source spend share, innovator-multiple-source spend share, average spend per prescription, NDC-level concentration, and the share of spending accounted for by the top ten NDCs. The category crosswalk's matched spending share rises from 26.6 percent in 2002 to 99.5 percent in 2024, averaging 79.4 percent across 2010–2024, so these measures are most informative in the later part of the panel.

3.3 Pool Membership Data

The first policy panel documents pool membership for all 50 states from 2002 through 2024. Because no reconciled public history exists (Section 1.4), I constructed the panel through three complementary approaches: systematic scraping of state Medicaid agency websites, including archived historical content accessed through the Wayback Machine; direct contact with each state's MDRP contacts by phone and email; and FOIA requests or state Public Records Requests for states with incomplete responses. The resulting timeline was validated against KFF State Health Facts, NASHP (Horvath, 2019), OIG (2014), and CMS supplemental-rebate-agreement tables.

Each contested state was resolved against archived contemporaneous sources. One state (Tennessee, a comparison-group member immaterial to the headline result) remains provisional pending an agency reply. Because a staggered difference-in-differences coefficient is a weighted average over event-time cells defined by the treatment dates, accurate entry and exit dating is essential. The crosswalk and a full state-by-state decision log are released with the panel so that every coding choice can be checked.

Pool indicators are mutually exclusive within a state-year: each state-year is coded SSDC, NMPI, TOP\$, or NoPool. Five states switch from a vendor pool into SSDC during the panel (Delaware, Kentucky, Vermont, West Virginia, and, after a no-pool gap, Pennsylvania). SSDC itself is absorbing (no state exits), while NMPI and TOP\$ membership reverses for several states.

3.4 Pharmacy-Benefit Structure Panel

A second state-year policy panel measures pharmacy-benefit structure directly. It contains 747 coded state-year spells across all 50 states and tracks seven features: uniform PDL status, pharmacy carve-out status, MCO formulary autonomy, PDL vendor, PBM vendor, rebate-negotiation model, and major reform flags. These are the features through which a pool's bargaining power is (or is not) converted into steerable volume, and they enter the analysis as a uniform sensitivity and mechanism check rather than as headline controls (Section 4.2).

The coding strategy is intentionally conservative and built at the spell level. Each spell records a start year, end year, source basis, certainty rating, and coder notes. Many spell boundaries are supported by official current manuals, archived state materials, and managed-care documents, while some remain medium-certainty backcasts from later official manuals rather than fully contemporaneous archived sources. An important limitation, central to the control policy below, is timing: the governance variables are densely coded only from 2010 onward (roughly 2 percent

non-missing before 2010, versus 92–100 percent from 2010). For the many pre-2010 pool entries, the panel cannot observe whether a governance reform preceded or followed pool entry.

3.5 Treatment Timing and Analytic Samples

SSDC adoption timing is central to the design. The fifteen SSDC adopters enter in eleven cohorts:

Cohort	States
2006	Iowa, Maine, Vermont
2007	Utah
2008	West Virginia, Wyoming
2009	Oregon
2012	Mississippi
2015	North Dakota
2016	Delaware
2017	Ohio, Oklahoma
2020	Pennsylvania
2023	South Dakota
2024	Kentucky

The lead dynamic sample focuses on the eight later adopters first treated in 2012 or later (Mississippi, North Dakota, Delaware, Ohio, Oklahoma, Pennsylvania, South Dakota, and Kentucky), compared against the thirteen never-in-any-pool states. Those thirteen comparison states are Alabama, Arizona, California, Colorado, Florida, Illinois, Indiana, Kansas, Massachusetts, Missouri, New Jersey, New Mexico, and Texas. They span all four Census regions and include four of the five most populous states (California, Texas, Florida, and Illinois) alongside small programs, so the counterfactual is a broad cross-section rather than an outlier fringe. This later-adopter restriction, not an asymmetric set of controls, is how the design handles the early cohorts’ truncated pre-periods (Section 4.1 explains why). For the reversing pools I use all treated states on the full sample.

Early adopters (2006–2009) were predominantly smaller, rural states with limited in-house pharmaceutical negotiating capacity. The 2012–2017 wave brought a mix of small and medium-

sized states, including Ohio and Oklahoma. The most recent adopters include Pennsylvania, one of the largest state Medicaid programs. This variation in adopter characteristics matters for interpreting generalizability.

3.6 Descriptive Statistics

Table 1 reports the supplemental rebate share by corrected pool family.

Table 1. Supplemental Rebate Share by Pool Family

Pool family	Mean	Median	SD	Min	Max	State-years
SSDC	5.32%	4.83%	4.28%	0.00%	28.40%	181
TOP\$	3.52%	3.61%	2.56%	−0.08%	12.35%	149
NMPI	3.11%	2.20%	3.56%	−8.11%	28.24%	202
NoPool	1.77%	0.16%	2.76%	−2.98%	21.83%	579

Source: Author's calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

SSDC state-years have the highest supplemental rebate percentages. The large mean-median gap for no-pool states (1.77 vs. 0.16 percent) reflects a right-skewed distribution with many states reporting near-zero supplemental rebates. These raw gaps are the motivation, not the finding: SSDC state-years also differ in governance and drug-mix profiles, which the design addresses through within-state staggered comparisons against never-pooled states and through the robustness checks below.

Table 2. Drug-Mix Descriptives by Pool Family (SDUD, 2010–2024)

Pool family	Single-source share	Innovator multi-source share	Avg. spend/Rx	Top-10 NDC share
NoPool	42.1%	21.6%	\$94.17	15.5%
SSDC	50.1%	21.4%	\$93.37	18.9%
NMPI	46.8%	20.5%	\$96.21	17.3%
TOP\$	43.4%	25.0%	\$94.64	16.6%

Source: Author's calculations from State Drug Utilization Data.

Drug baskets vary modestly across pool families. SSDC states carry a somewhat higher single-source (protected-brand) spend share, which motivates the SDUD-based composition checks and the class-substitutability test in Section 5.

Because the identifying contrast is later SSDC adopters versus never-pooled states, Table 2b reports baseline (2002–2005) characteristics by group. Two features matter for interpretation. First, the eight later adopters begin with *low* baseline rebate capture (0.30 percent of drug spending), below both the never-pooled states (1.29 percent) and the early SSDC adopters (1.26 percent). This is the negative-selection pattern examined in Section 5.1. Second, the never-pooled comparison states are on average somewhat larger (mean log drug spending 20.5 versus 19.6 for later adopters; Section 3.5). State fixed effects absorb fixed size differences, but the imbalance is one reason the later-adopter event study and the leave-one-comparison-state-out check (Section 5.4) matter. Baseline drug-mix shares are similar across groups, though the 2002–2005 SDUD crosswalk is thin (Section 3.2), so drug-mix balance is more reliably read on the post-2010 panel (Table 2).

Table 2b. Baseline (2002–2005) Characteristics by Group

Group	States	Supp. rebate (pp)	Mean log drug spend
Never-pooled	13	1.29	20.49
SSDC early (2006–2009)	7	1.26	18.83
Vendor (NMPI/TOP\$)	22	0.38	19.89
SSDC later (2012+)	8	0.30	19.58

Source: Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

4. Empirical Strategy

4.1 Identifying Variation

The design is a staggered difference-in-differences. States that joined SSDC are considered as the “treated” group, while the thirteen states that never joined any pool serve as “untreated” controls. The design compares the change in supplemental rebate capture within states that joined SSDC, before versus after joining, with the change over the same years in the never-pooled states. State fixed effects remove permanent differences across states, and year effects remove national shocks, such as the post-2010 escalation of federal statutory rebates, that hit all states together. What remains identifies the effect of joining, provided treated and comparison states would have trended similarly in the pool’s absence. This is the parallel-trends assumption, which the pre-entry data can probe but never prove. The credible SSDC cohort is the later adopters, meaning the eight states first treated in 2012 or later. The reason is mechanical, not substantive. The panel opens in 2002, and the six states that joined in 2006–2009 are treated almost from the start, leaving no untreated years in which to observe how they were trending before entry. An estimate that pools them in is dragged toward zero by construction, not by evidence, and Section 5.2 shows the two estimates side by side. The eight later adopters have long observed pre-periods.

Because pool entry can travel with other pharmacy-benefit reforms, I fix the target quantity in advance. The headline estimand is the *total effect* of joining SSDC as implemented (pool participation plus whatever co-timed changes travel with it) for later adopters, relative to never-pooled states. The vendor-pool and any-pool estimands are defined by the same rule on the full sample and are reported in Appendix B; no pool receives a hand-picked specification. A state-

by-state check (Section 5.4) then asks whether co-timed reforms could plausibly account for the headline.

4.2 Governance Variables as Bundled Treatments, Not Controls

A natural instinct is to add the governance variables (uniform PDL, carve-out, negotiation model) as regression controls and read the adjusted coefficient as the “clean” pool effect. There are two problems with that approach. Conceptually, several of these variables are the very channels through which pooling is supposed to work: states unify their PDLs or extend rebate collection *because* they pooled. Adjusting for a channel that the treatment itself moves (a “bad control”) can attenuate or even flip a real effect. Practically, the governance panel is densely coded only from 2010 onward, so the controlled specification silently drops roughly a third of the sample and is identified almost entirely on the post-2010 window. The control policy is therefore symmetric and fixed in advance: the uncontrolled total effect is primary for every pool, and the governance-controlled specification is a uniform sensitivity check, reported in Appendix D, never as anyone’s preferred estimate, and not interpretable as the causal direct effect of pooling. Where the bundling concern is real, the design addresses it at the level of named states and dated reforms (Section 5.4) rather than by regression adjustment.

4.3 Estimation and Inference

Staggered adoption breaks the classic two-way fixed-effects regression, which implicitly compares newly treated states to already-treated ones and can badly distort estimates when effects grow over time. I therefore use estimators built for staggered timing, matched to each pool’s institutional facts. No state ever leaves SSDC, so SSDC adoption is an absorbing treatment. The primary estimator (Sun and Abraham, 2021) compares each adopting cohort only to never-pooled states, year by year around entry, and averages the post-adoption effects over the

first nine event years, the horizon the data can support (Section 5.3). I corroborate that average with four alternative estimators that make different technical choices (Callaway and Sant’Anna, 2021; Borusyak, Jaravel, and Spiess, 2024; de Chaisemartin and D’Haultfoeuille, 2020; and a fully saturated event study) and report the spread across implementations rather than a single number. States enter *and exit* the vendor pools, so those estimates use the de Chaisemartin-D’Haultfoeuille estimator, which is designed for treatments that switch off. The estimating equation and estimator mechanics are collected in Appendix G.

Inference deserves separate attention because it disciplines the claims throughout the paper. Only eight treated states identify the SSDC estimate, and conventional clustered standard errors are known to overstate significance with so few clusters. All headline inference therefore uses the wild cluster bootstrap, a resampling method built for exactly this situation, and I report its p-values exactly. Where an analytic p-value appears, it is flagged as overstating significance. Two complements probe the same worry from different directions: randomization inference over the eight adoption dates (Section 5.2), and the Rambachan and Roth (2023) honest-bounds framework, which asks how badly the parallel-trends assumption would have to fail before the conclusion changes. Appendix G details all of these, along with the two-way fixed-effects benchmark and its Goodman-Bacon (2021) diagnosis. The estimand, sample, treatment definition, fixed effects, clustering, and planned robustness checks were all fixed in advance of estimation, and the headline estimate was reproduced in an independent software implementation. The robustness battery is reported alongside the results in Section 5.

***## 5. Results

5.1 Descriptive Evidence: Trends, Timing, and Selection into Pools

It is useful to start with what the raw data show. Figure 3 plots mean supplemental rebate capture over calendar time by pool family. SSDC states trend above no-pool states throughout - SSDC state-years average 5.3 percent of drug spending against 1.8 percent for never-pooled state-years (Table 1) - and the gap widens after the general rise in supplemental rebates post-2020, with NMPI and TOP\$ tracking in between. Figure 4 shows when the treatment arrives. SSDC entry is staggered across eleven cohorts from 2006 to 2024, with thin support at late event times (Section 5.3).

Raw gaps like these are where the policy enthusiasm for pooling comes from, but on their own they could simply mean that high-capture states join pools. The pre-entry record says otherwise: the selection runs in the opposite direction. In the years *before* they adopted SSDC, the eight later adopters captured a supplemental rebate share of 1.33 percent of drug spending, below the 2.31 percent of the never-pooled comparison states and the 1.74 percent of states that were in a vendor pool but never joined SSDC. The same ordering holds in the 2002–2005 baseline (Table 2b). States appear to have turned to SSDC because their existing rebate capture was inadequate, not because they were already extracting more. That makes the post-entry catch-up visible in Figure 3 more interesting, not less. However, negative selection is descriptive context rather than identification evidence: low baseline capture can also predict mean reversion or coincide with other reforms, which is what the design (Section 4.1) and the co-timed reform check (Section 5.4) address.

5.2 The Causal Estimate: SSDC Later Adopters

Table 3 reports the central estimate, which is the formal version of the catch-up pattern visible in Figure 3. Later adopters saw large gains in rebate capture after joining: in the later-adopter

sample (first treated 2012 or later), the average post-adoption effect on the supplemental rebate share is **3.38 percentage points** (Sun-Abraham), with a wild cluster bootstrap p-value of 0.051. This is just above the conventional 0.05 threshold, and I report it exactly rather than round it down. For scale, the never-pooled base is under 2 percent of drug spending.

Table 3. SSDC Later-Adopter Effect and Robustness (Uncontrolled)

Check	Result
Headline (Sun-Abraham, later adopters, uncontrolled)	3.38 pp , WCB p = 0.051
Cross-estimator	SA 3.38 / CS 2.66 / BJS 4.27 / dCDH ~2 to 3.8 / saturated 3.66 (all positive)
Leave-one-state-out	[2.91, 3.88] pp - not driven by one state
Outliers / winsorize 1/99	3.07–3.38 pp
Outcome-construction covariates (headroom, drug-mix, post-federal denominator)	3.6–4.0 pp, WCB 0.03–0.04 (sharpens)
Rambachan-Roth honest bounds	survives under exactly parallel trends [1.11, 5.49]; breaks under modest trend violations
Governance-controlled (sensitivity specification only)	sign-unstable / imprecise; not headline-grade
One feature deserves emphasis: the estimate <i>sharpens</i> rather than attenuates when the valid	

outcome-construction covariates (which adjust for what a state could mechanically capture given the federal floor and its drug mix) are added. The governance-controlled specification, by contrast, is sign-unstable and wide, exactly as Section 4.2 predicts (Appendix D).

Which SSDC estimand is credible matters, and Table 3b shows why, side by side. The all-treated SSDC estimate, which pools in the six early adopters, is a null 0.6 pp - the mechanical truncation described in Section 4.1, not evidence against an effect. Restricting to later adopters gives 3.38 pp, and extending the outcome panel back to 1997 (Appendix B) gives 3.34 pp on essentially the same sample. The credible SSDC estimate is for later adopters, and I label it as such throughout.

Table 3b. SSDC Estimates by Sample

Sample	Estimator	Effect (pp)	Note
All SSDC adopters (15), 2002–2024	Sun-Abraham	0.6	null; early cohorts have truncated pre-

Sample	Estimator	Effect (pp)	Note
Later adopters (8), 2002–2024	Sun-Abraham	3.38	periods headline (WCB p = 0.051)
Later adopters (8), 1997–2024 extended panel	Sun-Abraham	3.34	stable to a longer pre-period

Source: Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Randomization inference asks how often a placebo “effect” as large as the real one appears when the eight adoption dates are handed out at random to states in the analytic sample. Almost never.

The placebo distribution is centered at zero (mean 0.04 pp, SD 1.20, with a 95th percentile of absolute placebo effects of 2.34 pp), and the real 3.38 pp lies far in its tail, with a two-sided randomization-inference $p = 0.004$. This design-based test does not rely on small-cluster approximations. The headline estimate weights states equally and so describes the typical later adopter. Weighting by drug spending, which is the fiscally relevant quantity, gives a larger 4.37 pp, because the gains are larger among the bigger recent adopters. Treatment-timing choices do not move the result: lagging treatment by one year gives 3.52 pp, and dropping the partial entry year gives 3.98 pp (Section 5.3 reports the dynamic ramp).

One multiple-testing caveat is worth stating plainly. The paper reports five pool definitions, and it is fair to ask whether the borderline SSDC p -value survives family-wise correction. Across the pool-family tests, Holm adjustment leaves only any-pool near significance (adjusted $p = 0.065$) and inflates the individual SSDC test to 0.204; Benjamini-Hochberg control leaves SSDC, TOP\$, Prime, and any-pool at adjusted p -values of 0.06–0.09. The SSDC later-adopter result is therefore best defended on its own design-based evidence (the randomization-inference $p = 0.004$, the robustness battery, and the clean-entry and dynamic-ramp patterns), not on a single

cross-pool p-value. I treat the cross-pool ranking as suggestive rather than as five independent confirmed effects.

5.3 Dynamics and Cohort Support

If pooling genuinely works, a large same-year jump in rebate capture would be surprising. States renegotiate class by class through contract renewals, PDL reviews, and prior-authorization updates, which implies a one-to-three-year ramp rather than an overnight change. That is what the data show. Figure 5 plots the SSDC event study. Pre-treatment leads are flat and individually insignificant, consistent with the de Chaisemartin-D'Haultfoeuille placebos. The post-adoption path rises gradually and peaks around event years three to five. The ramp is quantitative, not just visual: aggregating the dynamic coefficients by horizon, the effect is essentially zero in the entry window (event times 0–1: 0.2 pp), reaches 3.2 pp by years 2–3, and 5.1 pp by years 4–8. A jump at entry would more plausibly signal a mechanical reporting change than a bargaining response. I truncate the plotted path at event time 8. Event times of nine or more are identified by three or fewer states, and eleven or more by a single state (North Dakota), so the far-right tail reflects which one or two states happen to be observed rather than dynamic structure. Inference rests on the aggregated effect, not on individual late-event-time coefficients.

5.4 Co-Timed Reforms and Confounding

The central threat to a causal reading is that SSDC entry coincides with other pharmacy-benefit reforms (a uniform PDL binding managed care, a carve-out, the extension of supplemental rebate collection to managed-care utilization, or a single state PBM), each of which can independently move the rebate share. To make that bundling checkable rather than asserted, Table 4 records, for each of the eight later adopters, the year of every such reform relative to SSDC entry, coded from the primary-source reform record. The pattern is reassuring for identification. Only two of

the eight, Delaware (2016) and Pennsylvania (2020), entered SSDC in the same year as a major reform; the other six entered without one, and where reforms occur they typically arrive three or more years after pool entry or several years before it (Table 4).

Table 4. Policy-Bundle Review: Co-Timed Reforms Around SSDC Entry

State	SSDC entry	Uniform PDL (MCO)	Carve-out	MCO rebate collection	Single PBM
Mississippi	2012	-	-	post (+2)	post (+12)
North Dakota	2015	-	post (+5)	post (+2)	-
Delaware	2016	-	-	same-yr	-
Ohio	2017	post (+3)	-	post (+3)	post (+5)
Oklahoma	2017	-	-	post (+7)	-
Pennsylvania	2020	same-yr	-	same-yr	-
South Dakota	2023	-	-	-	-
Kentucky	2024	pre (2021)	-	pre (2021)	pre (2021)

Source: Author's coding of pharmacy-benefit reforms from primary-source state Medicaid

materials.

The corresponding estimate confirms the review. Re-estimating the later-adopter Sun-Abraham effect on the clean-entry subsample, the six states with no major reform within one year of SSDC entry (dropping Delaware and Pennsylvania), yields 3.11 pp (SE 0.88), essentially the full-sample 3.38 pp. Dropping the three largest recent adopters (Pennsylvania, Ohio, Kentucky) instead gives 3.61 pp. The headline is therefore not an artifact of pool entry coinciding with managed-care rebate collection or formulary centralization; it is present in the states that joined SSDC *without* a simultaneous governance change. This does not eliminate the bundling concern, as the estimand remains pool participation as implemented and unobserved co-timed changes cannot be ruled out, but it substantially narrows it. (Appendix D reports the governance-controlled sensitivity specification.)

Two further checks address the small comparison group directly. First, leave-one-comparison-state-out: dropping each of the 13 never-pooled states in turn leaves the SSDC estimate in [3.12,

3.60] pp (lowest when California is dropped, highest when Arizona is), so no single comparison state drives the result. Second, an alternative donor pool: replacing the never-pooled comparison group with the states that were ever in a vendor pool but never in SSDC, a completely different counterfactual, yields 3.41 pp, essentially the headline. The estimate is thus not an artifact of the particular 13-state comparison group, though that group's small size remains the binding constraint on precision (Section 6.3).

5.5 Vendor-Administered Pools

The gains documented above arrived primarily through the SSDC; the two vendor-administered pools do not share its pattern. Estimated with an estimator that accommodates their entry-and-exit membership, NMPI raises the supplemental rebate share by 1.90 percentage points (bootstrapped CI [0.0, 3.8]) and TOP\$ by 0.34 points with a confidence interval spanning zero (CI [-1.0, 1.7]); pooling of any kind averages 1.22 points (CI [0.1, 2.3]). The TOP\$ interval rules out effects as large as the SSDC later-adopter estimate, but modest positive or negative effects remain possible, and a direct SSDC-versus-other-pools contrast (about +3.2 pp) is not statistically distinct under small-cluster inference. The ordering (the state-owned SSDC at least matching the vendor pools, with the weakest vendor pool a null) is consistent with the incentive-alignment and steerable-volume mechanisms of Section 2.4, though pool type is chosen, not assigned, and selection cannot be ruled out. Appendix B reports the full multi-pool results table, the estimator detail, and further institutional background on NMPI and TOP\$.

5.6 Theory-Motivated Heterogeneity

Table 5 collects the heterogeneity cuts. They are directionally consistent with the bargaining theory but imprecise under small-cluster inference, and one is non-confirmatory.

Table 5. Heterogeneity (Theory-Motivated; Suggestive, Underpowered)

Cut	Result	Theory prediction
Small vs. large states (SSDC)	small +3.37 vs. large +1.25 pp; WCB p = 0.33	larger for small: yes (directional)
Small vs. large (any-pool)	small +2.52 vs. large +1.02 pp; WCB p = 0.14	yes (directional)
Switchers-in vs. direct (SSDC)	2.37 vs. 2.78 pp; WCB p = 0.89	no “already-captured” discount
Pool type	SSDC/NMPI positive; TOP\$ null	state-owned/credible > vendor: yes
Class substitutability (state-year proxy)	non-confirmatory; leans opposite, ns	larger where substitutes exist - not supported
Smaller states made the largest gains, about twice those of larger states, under both SSDC and		

any-pool. This is the sign the countervailing-power model predicts, since pooling relaxes the small-state bargaining constraint most, but the differential is not statistically distinguishable from zero. States that switched into SSDC from a vendor pool gain about as much as direct adopters, which provides no support for an “already-captured-gains” discount.

The class-substitutability prediction is the one cut that does not confirm. Using a state-year drug-mix proxy for substitutability (the innovator-multiple-source spend share, and one minus the single-source share), the pool effect is no larger, and if anything mildly smaller, in more-substitutable states, and every differential is insignificant under the wild cluster bootstrap. I report this honestly as non-confirmatory rather than omit it. Two caveats limit what it can establish. The proxy is a state-year drug mix, not a within-class measure, so it confounds “this state buys more substitutable drugs” with correlated state characteristics. And a coherent alternative reading of the (insignificant) sign is that supplemental rebates are the only price lever on protected sole-source drugs, so pooling’s surplus may concentrate where spend is in those high-margin brands. A genuine test of the class-level mechanism requires an SDUD-by-therapeutic-class panel, which I flag as future work. I do not treat class substitutability as supporting evidence.

5.7 Secondary Outcomes, Steering, and the Capture-Savings Distinction

The primary outcome is rebate *capture*, not net spending, and Table 6 makes the distinction concrete by re-running the later-adopter Sun-Abraham estimator on other outcomes built from the same CMS-64 and SDUD data. Three patterns stand out.

First, capture is not demonstrated savings, but neither is it offset. Pooling raises the supplemental rebate share by 3.4 pp and supplemental rebate dollars per prescription by \$3.19, while log gross drug spending does not move and log drug spending net of supplemental rebates is directionally lower but not significant (Table 6). So the captured rebate is not clawed back through higher gross outlays - there is no evidence of the high-list-price, high-rebate branded channeling that would make capture illusory. However, the net-spending effect is too imprecise to assert a savings, and the design measures neither welfare nor manufacturer pricing responses. The paper therefore reserves the word “savings.” I net out only the *supplemental* rebate: the federally set national rebate share trends up over the panel for reasons unrelated to pooling, and substituting the directly reported CMS-64 net series (gross minus *all* rebates) produces a spurious -37.8 pp (SE 7.2) “savings” that measures federal rebate policy, not pooling (Appendix J).

Second, the rebate gain is not a mechanical drug-mix artifact. The single-source (protected-brand) and innovator-multiple-source spend shares do not move (Table 6), so pooling is not simply shifting states into higher-rebate drug categories.

Third, and most informative for mechanism, pooling concentrates utilization. The within-state top-10-NDC spend share rises 2.2 pp (SE 0.69) and the NDC-spend Herfindahl rises. This is the formulary-steering channel the theory predicts: a credible pool steers volume onto the preferred (rebated) products - direct, positively-signed mechanism evidence, in contrast to the non-confirmatory substitutability proxy of Section 5.6.

Table 6. Secondary Outcomes (SSDC Later-Adopter Sun-Abraham)

Outcome	Effect	SE
Supplemental rebate share (pp) - <i>primary</i>	3.38	0.77
Log gross drug spending	-0.04	0.05
Log drug spending net of supplemental rebates	-0.06	0.06
Supplemental rebate \$ per prescription	3.19	1.04
Single-source spend share (pp)	-0.03	0.66
Innovator-multi spend share (pp)	-0.83	0.85
Top-10 NDC spend share (pp) - <i>steering</i>	2.18	0.69

Source: Author's calculations from CMS-64 Financial Management Report and State Drug

Utilization Data. "Net of supplemental rebates" is gross spending times one minus the supplemental rebate share. The federally set national rebate is deliberately left in (see Appendix J). Enrollment-denominated per-capita outcomes are unavailable in the CMS-64 and are not estimated.

It is worth being concrete about why the sharpest possible mechanism test - the pool effect on rebate capture *within* therapeutic class interacted with class-level substitutability - is infeasible with these data. Three reasons. First, supplemental rebates are reported only as a single state-level line in the CMS-64; they are never broken out by drug or therapeutic class, so class-level rebate capture is unobservable regardless of any crosswalk. Second, no NDC-to-therapeutic-class (ATC) crosswalk is available in these data; the only class information is the CMS single-source / innovator-multiple-source / non-innovator category, which is a coarse substitutability proxy. Third, the NDC-level SDUD needed to build any within-class utilization measure covers 2011–2016, which predates the entry of six of the eight later adopters. The feasible mechanism evidence is therefore the state-year steering result above. A true within-class design would

require class-coded rebate data (for example, T-MSIS under a data-use agreement) joined to an ATC crosswalk.

6. Discussion

6.1 Interpretation and Policy Implications

Later adopters saw large gains in rebate capture after joining SSDC. States that later joined SSDC were capturing less than their never-pooled peers before joining, and their capture rises steadily after entry: joining SSDC raised the supplemental rebate share of drug spending by about 3.4 percentage points among the eight later adopters, and the estimate is stable across five estimators, leave-one-state-out checks, and a state-by-state check of co-timed pharmacy-benefit reforms. Smaller states made the largest gains, and the gains arrived primarily through the SSDC; averaged across all pool types, the gain is a more modest one to two points. Against a no-pool base of under two percent and a binding federal rebate floor, a few-percentage-point gain is a real fiscal lever, but a modest one, not a transformation. The policy reading is correspondingly measured: pooling may be a useful fiscal tool when paired with governance arrangements that make volume steering credible, but nominal pool membership alone should not be interpreted as sufficient to reduce net drug spending. Pooling is especially relevant for smaller states, but it will not, by itself, close the gap between low-capture and high-capture states.

The pool-type pattern carries the sharpest policy content. The state-owned SSDC and NMPI deliver positive effects while TOP\$ does not, and the ordering aligns with how credibly each arrangement converts pooled lives into a disagreement threat and with whose budget the bargaining serves. For a state choosing whether and how to pool, the design of the pool (alignment between who bargains and who keeps the savings, state-specific contracting that

preserves price discrimination, a repeated member-state governance cycle) plausibly matters as much as the decision to pool at all. That said, SSDC states differ in ways (unified PDLs, MCO rebate collection, recent large joiners) that the design cannot fully separate from governance; the SSDC estimate is best read as the effect of SSDC as implemented, bundle included.

The federal financing environment gives these magnitudes their current policy weight: the 2025 reconciliation law is projected to withdraw roughly \$911 billion in federal Medicaid financing over the coming decade (Euhus et al., 2025), and pooled rebate negotiation is one of the few levers that extracts resources from manufacturers rather than cutting coverage or payment (Section 1.1). The estimates discipline expectations about how much that lever can carry: a meaningful offset to drug-spending pressure, and among the least painful levers available, but not remotely a substitute for the federal financing being withdrawn.

To fix orders of magnitude (not to forecast a budget line), consider the scale. Gross Medicaid drug spending is roughly \$80 billion, of which the fifteen SSDC states account for about \$20 billion, leaving some \$60 billion outside SSDC. Applying the any-pool central estimate (1.2 pp) to that \$60 billion implies on the order of \$0.7 billion in additional supplemental rebates a year; applying the larger SSDC later-adopter estimate (3.4 pp) implies closer to \$2.0 billion. Three forces pull the realistic figure toward the lower end and possibly below it. The estimates are imprecise and several intervals include zero. The gains are not separable from the governance reforms that travel with pooling. And as adoption saturates, the marginal state has less untapped leverage to capture. These projections therefore bound a plausible range, roughly \$0.4–1.0 billion annually under conservative assumptions, and should be read as an upper-bounded order of magnitude, not an expected return on nominal pool membership.

6.2 Relation to Prior Literature

The finding sits naturally in the buyer-power and pooled-procurement literatures. Chown et al. (2019) and Ho and Lee (2024) imply that scale raises rebates only when attached to a credible formulary decision; the small-state and pool-type heterogeneity here is the Medicaid-specific version of that prediction. The international evidence (Dubois et al., 2021; Parmaksiz et al., 2022) that pool design drives outcomes is mirrored in the SSDC-versus-TOP\$ contrast. And the pharmacy-benefit literature (Dranove et al., 2021; Hernandez and Gellad, 2020) explains why governance is part of the mechanism rather than a nuisance to be adjusted away.

6.3 Limitations

Potential limitations include threats to identification, measurement error, external validity, and policy interpretation, which I take in turn.

Identification. Even though the observed pre-treatment coefficients are flat (Section 5.3), differential trends cannot be ruled out, and the Rambachan-Roth bounds make this concrete: the SSDC effect survives the assumption of exactly parallel trends but breaks under modest violations. Pool entry can co-time with other pharmacy-benefit reforms (uniform PDLs, carve-outs, vendor or negotiation-model changes), so the estimate reflects pool participation as implemented rather than pooling in isolation. Because the later adopters started from below-average baseline capture, mean reversion among low-baseline states is a live alternative to a genuine pooling effect. Adoption is policy-endogenous (states choose whether and when to pool), and the timing of treatment is itself measured with some uncertainty for a few states, which matters for a staggered design (Section 3.3). Lastly, cross-state or cross-manufacturer spillovers would violate SUTVA: if a manufacturer's concession to a pool changes the terms it offers non-pool states, the never-pooled comparison group is not a pure counterfactual.

Measurement. The CMS-64 reports on a fiscal-year basis, which need not align with calendar-year pool-membership indicators. I do not anticipate this misalignment biasing the estimates, as pool membership changes typically take effect at the beginning of a state fiscal year, and re-estimating with treatment lagged one year (3.52 pp) or the partial entry year dropped (3.98 pp) leaves the headline intact. The negative and missing outcome values documented in Section 3.1 are a further concern; to limit their influence, I 1) validate the CMS-64 outcome against SDUD, and 2) re-estimate with outliers dropped or winsorized, which leaves the estimate at 3.07–3.38 points. The outcome is a rebate *share*, not a rebate *rate* negotiated per unit, so composition changes can move it independently of bargaining. Most important for interpretation, managed-care rebate-reporting changes can mechanically move the numerator: a state that begins collecting and reporting supplemental rebates on managed-care utilization raises the measured share without necessarily negotiating harder. And the SDUD category crosswalk is weak in the early years (matched spending share is low before 2010), so the drug-mix measures are most informative in the later part of the panel.

External validity. The credible SSDC estimate is identified off later adopters, who differ from the early adopters in size and capacity, so the headline does not necessarily describe the early cohorts. The remaining non-adopters may not resemble prior adopters, which limits how well the estimates generalize to the future marginal adopter. The pattern documented here should therefore be read as the effect for states like the later adopters, not as a universal pooling effect.

Policy interpretation. Rebate capture is not net drug spending: a rebate-maximizing formulary can steer utilization toward high-rebate branded drugs, so realized rebate dollars need not translate one-for-one into lower net spending. State fiscal savings are smaller still than total rebate dollars, because the federal government shares in Medicaid drug costs through FMAP, so

a captured rebate is split between state and federal budgets. And the access and prescribing consequences of pool-negotiated formularies are not measured here.

One cross-cutting point closes the discussion: the binding constraint on precision is structural. By 2024 most states had pooled, leaving only thirteen never-in-any-pool comparison states and between eight and a few dozen treated clusters depending on the pool, so several confidence intervals hug zero and the heterogeneity cuts are underpowered. This imprecision cannot be fixed by adding already-treated states, as the comparison group is the scarce resource, but it can be mitigated by longer post-periods as recent adopters accumulate data, by better pre-period data, by alternative donor designs, and by class-level outcomes.

6.4 Future Research

The most valuable extension is a class-level mechanism design; Section 5.7 details why it cannot be built with these data. Short of that, the state-year steering evidence in Section 5.7 (pooling concentrates utilization onto the top NDCs) is the strongest available mechanism signal and points the way. A second priority is the net-spending question. This paper finds no detectable change in net drug spending alongside the capture gain, but a design powered to detect modest net-spending effects, and to test whether rebate gains are partly offset by branded-drug channeling, as Munshi et al. (2018) and Hernandez and Gellad (2020) would suggest, would sharpen the capture-versus-savings distinction. Lastly, as more recent large adopters accumulate post-period data, the precision constraint will ease, and the pool-type and size heterogeneity that are only directional here may become testable.

7. Conclusion

Two-thirds of U.S. states have joined an interstate Medicaid drug purchasing pool, yet the question of whether pooling actually raises supplemental rebate capture has rested on anecdote. Using a corrected, primary-source panel of pool membership for all 50 states from 2002 through 2024 and estimators matched to each pool's institutional structure, this paper finds that joining the state-owned SSDC modestly raises supplemental rebate capture (roughly three percentage points among later adopters, with larger gains for smaller states), while the vendor-administered pools show smaller or statistically indistinguishable effects and pooling averages one to two points overall. The pattern is consistent with a countervailing-buyer-power mechanism in which pooling helps where the pool can credibly steer volume and where the party that keeps the savings governs the bargaining. Under the maintained difference-in-differences assumptions, the evidence is consistent with a modest positive effect. The estimates are imprecise, for the structural reason that most states have already pooled, and they remain sensitive to modest departures from parallel trends.

The lesson for states is measured: pooling can be a useful fiscal tool where governance makes volume steering credible, but nominal membership alone is not sufficient to reduce net drug spending. In a decade when federal law is withdrawing hundreds of billions of dollars in Medicaid financing, that measured conclusion still matters: pooled rebate negotiation is one of the few cost-containment levers that asks manufacturers, rather than enrollees or providers, to absorb part of the adjustment.

***## References

Bendicksen, L., and Kesselheim, A.S. (2022). Anticipated efficiencies, real costs: Medicaid managed care organizations and the pharmacy benefit. *Journal of Managed Care & Specialty Pharmacy*, 28(3), 354–361.

Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6), 3253–3285.

Callaway, B., and Sant’Anna, P.H.C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.

Cameron, A.C., Gelbach, J.B., and Miller, D.L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.

Chown, J., Dranove, D., Garthwaite, C., and Keener, J. (2019). The opportunities and limitations of monopsony power in healthcare: Evidence from the United States and Canada. NBER Working Paper 26122.

de Chaisemartin, C., and D’Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9), 2964–2996.

Dranove, D., Ody, C., and Starc, A. (2021). A dose of managed care: Controlling drug spending in Medicaid. *American Economic Journal: Applied Economics*, 13(1), 170–197.

Dubois, P., Lefouili, Y., and Straub, S. (2021). Pooled procurement of drugs in low and middle income countries. *European Economic Review*, 132, 103655.

Duggan, M., and Scott Morton, F. (2006). The distortionary effects of government procurement: Evidence from Medicaid prescription drug purchasing. *Quarterly Journal of Economics*, 121(1), 1–30.

Euhus, R., Williams, E., Burns, A., and Rudowitz, R. (2025). Allocating CBO's estimates of federal Medicaid spending reductions across the states: Enacted reconciliation package. KFF, July 2025.

Gifford, K., Winter, A., Wiant, L., Dolan, R., Tian, M., and Garfield, R. (2020). How state Medicaid programs are managing prescription drug costs: Results from a state Medicaid pharmacy survey for state fiscal years 2019 and 2020. KFF and Health Management Associates, April 2020.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277.

Hernandez, I., and Gellad, W.F. (2020). Differences between managed care and fee-for-service Medicaid in the use of generics for high-rebate drugs: The cases of insulin glargine and glatiramer. *Journal of Managed Care & Specialty Pharmacy*, 26(2), 154–159.

Hinton, E., Williams, E., Raphael, J., Mudumala, A., Rudowitz, R., Gifford, K., Lashbrook, A., and Rosenzweig, C. (2025). A view of Medicaid today and a look ahead: Balancing access, budgets and upcoming changes. Results from the 25th annual Medicaid budget survey. KFF and Health Management Associates, November 2025.

Ho, K., and Lee, R.S. (2024). Contracting over pharmaceutical formularies and rebates. NBER Working Paper 32790.

Horvath, J. (2019). State initiatives using purchasing power to achieve drug cost containment. National Academy for State Health Policy.

KFF (2025). Health provisions in the 2025 federal budget reconciliation law. KFF, 2025.

MacKinnon, J.G., and Webb, M.D. (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2), 233–254.

Medicaid and CHIP Payment and Access Commission (2018). Medicaid payment for outpatient prescription drugs. MACPAC Issue Brief, May 2018.

Munshi, K.D., Mager, D., Ward, K.M., Mischel, B., and Henderson, R.R. (2018). The effect of Florida Medicaid's state-mandated formulary provision on prescription drug use and health plan costs in a Medicaid managed care plan. *Journal of Managed Care & Specialty Pharmacy*, 24(2), 124–131.

Office of Inspector General, U.S. Department of Health and Human Services (2014). States' collection of offset and supplemental Medicaid rebates. Report OEI-03-12-00520, December 2014.

Park, C. (2022). Trends in Medicaid drug spending and rebates. MACPAC, October 2022.

Parmaksiz, K., Pisani, E., Bal, R., and Kok, M.O. (2022). A systematic review of pooled procurement of medicines and vaccines: Identifying elements of success. *Globalization and Health*, 18(1), 59.

Rambachan, A., and Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5), 2555–2591.

Sun, L., and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175–199.

Virabhak, S., and Shinogle, J.A. (2005). Physicians' prescribing responses to a restricted formulary: The impact of Medicaid preferred drug lists in Illinois and Louisiana. *American Journal of Managed Care*, 11(Spec No.), SP14–SP20.

Appendix

All results in this appendix use the primary-source pool-membership panel, 2002–2024, with the never-in-any-pool states (13) as the comparison group unless noted. Inference is wild cluster bootstrap throughout (mechanics in Appendix G3); analytic p-values overstate significance with so few treated clusters and are flagged where reported. The primary specification is the uncontrolled total effect, applied symmetrically to every pool.

Appendix A. Pharmacy-Benefit Variables

Variable	Description	Why it matters
Uniform PDL	State has a uniform preferred drug list spanning relevant delivery systems	Can increase formulary leverage; part of how a pool converts size into steerable volume
Pharmacy carve-out	Pharmacy benefit carved out of managed care	Changes who controls the rebate and utilization-management structure
MCO formulary autonomy	Degree of managed-care formulary independence	Captures whether managed-care plans dilute state leverage
PDL vendor	PDL support vendor or administrator	Administrative capacity and contracting style; sparse, descriptive only
PBM vendor	Pharmacy benefit manager or analogous contractor	Negotiation execution and reporting; sparse, descriptive only
Rebate-negotiation model	State-direct, SSDC-individual, collective-pool, PBM-assisted, hybrid, or unknown	Distinguishes pool membership from how bargaining is actually done
Major reform flag	Major administrative or policy transition in a spell	Flags reform-dense windows

These variables enter only as a uniform governance-controlled sensitivity specification (Section 4.2; Appendix D), never as the headline.

Appendix B. Multi-Pool Results and Master Robustness

This appendix collects the vendor-pool and any-pool results summarized in Section 5.5, institutional background on the vendor pools, and the pre-2002 panel extension.

B0. Vendor-pool institutional detail. The two vendor-administered pools negotiate collective supplemental rebate agreements (SRAs) on behalf of member states (founding years and membership counts in Sections 1.3 and 2.3). NMPI is administered by Magellan/Prime Therapeutics: the pool entity contracts directly with manufacturers, while member states choose which negotiated drugs to place on their own formularies; its membership includes large states such as New York and Michigan. TOP\$ is likewise vendor-administered (Prime/Provider Synergies). By contrast, SSDC members (Section 2.3) join through a consortium agreement cost-shared through Vermont (the managing member), without needing to contract with the consortium’s hired contractor (Optum).

B1. Main results by pool - primary uncontrolled total effect, vs. no-pool.

Treatment	Estimator	Effect (pp)	WCB CI / p	Treated	Comparison	Pre-trends
SSDC (later)	Sun- Abraham	3.38	WCB p = 0.051	8	13	clean
NMPI	dCDH (exit-robust)	1.90	CI [0.0, 3.8]	17	13	clean
TOP\$	dCDH	0.34	CI [-1.0, 1.7]	11	13	clean
Prime (NMPI or TOP\$)	dCDH	1.36	CI [0.15, 2.6]	27	13	clean
Any pool	dCDH	1.22	CI [0.1, 2.3]	37	13	clean

Notes: The comparison group is the same 13 never-in-any-pool states throughout (Section 3.5);

“Treated” is the number of states ever in each pool over 2002–2024, and the SSDC row is the eight later adopters, the headline sample. Wild cluster bootstrap inference is authoritative (Appendix G3); dCDH confidence intervals are bootstrapped. The SSDC headline’s robustness ranges appear in main-text Table 3. “Pre-trends: clean” means the dCDH placebo (pre-period) estimates are jointly small and statistically indistinguishable from zero.

Section 5.5 discusses these results; the SSDC all-treated estimate is null (Table 3b).

B2. Pre-2002 panel extension (uncontrolled). The cross-estimator and robustness rows for the SSDC headline appear in main-text Tables 3 and 3b; this note documents the pre-2002 extension. The pre-2002 extension parses CMS-64 Line 7A2 (“Drug Rebate Offset - State Sidebar Agreement,” defined on the CMS-64 form in 1998) to build a 1997–2024 outcome panel. The 1997–2001 years are pure pre-treatment: no pool membership is recorded before 2002, and every SSDC adoption is 2006 or later, so the extension lengthens the pre-period without adding treated mass. The later-adopter Sun-Abraham average post-adoption effect is essentially unchanged (3.34 pp vs. 3.38 pp), confirming the headline is not an artifact of the 2002 panel start.

Appendix D. Governance-Controlled Sensitivity Specification

Re-estimating each design with time-varying governance controls (uniform PDL, pharmacy carve-out, and the major reform flag) yields sign-unstable, imprecise estimates across pools - the expected symptom of conditioning on a part-mediator on a restricted post-2010 sample (Section 4.2). It is reported only as a controlled residual association.

Appendix F. Heterogeneity

The full heterogeneity table, with exact wild cluster bootstrap p-values, appears as main-text Table 5.

Class-substitutability test (non-confirmatory). Mirroring the size test (baseline 2002–05 median split; wild cluster bootstrap on the high-vs-low differential; uncontrolled), using the innovator-multiple-source spend share (primary) and one minus the single-source share (alternative) as state-year substitutability proxies:

Treatment	Proxy	High–low differential	WCB p
SSDC later (2012 or later)	innovator multi-source	–4.59 pp	0.13
SSDC all-treated	innovator multi-	–3.03 pp	0.13

Treatment	Proxy	High–low differential	WCB p
Any pool	source innovator multi-source	–1.15 pp	0.30
SSDC / any	1 – single-source	–0.9 to –0.02 pp	0.74–0.98
Section 5.6 discusses the caveats and the non-confirmatory reading of this cut.			

Appendix G. Estimation and Inference Details

This appendix collects the estimator and inference mechanics summarized in plain language in Section 4.3 of the main text.

G1. Primary estimator (SSDC). SSDC adoption is absorbing (Section 3.3) and is estimated with the Sun and Abraham (2021) interaction-weighted event study on the later-adopter cohorts against never-pooled states. The estimating equation is

$$Y_{st} = \alpha_s + \gamma_t + \sum_{k \neq -1} \beta_k D_{st}^k + \varepsilon_{st}$$

where Y_{st} = the supplemental rebate share for state s in year t , α_s and γ_t = state and year fixed effects, D_{st}^k = an indicator that a later-treated state is k years from its first SSDC year with $k = -1$ the omitted reference, and ε_{st} represents an error term. The sample is the eight SSDC states first treated in 2012 or later plus the thirteen never-pooled comparison states, and the summary parameter is the average post-adoption effect over event times 0 to 8 (Section 5.3). The average is corroborated with the estimators of Section 4.3, reporting the spread across implementations rather than a single number (B2).

G2. Exit-robust estimator (vendor pools). Because states enter and exit NMPI and TOP\$, the estimates for NMPI, TOP\$, the combined Prime pools (NMPI or TOP\$), and any-pool use the de Chaisemartin and D’Haultfoeuille (2020) dynamic estimator on the full sample against never-pooled states, reporting the average effect, its bootstrapped confidence interval, and the placebo

(pre-trend) estimates. Two-way fixed effects, where shown, is a benchmark only and is diagnosed with the Goodman-Bacon (2021) decomposition.

G3. Wild cluster bootstrap. All headline inference uses the wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008; MacKinnon and Webb, 2017) with Rademacher weights, 9,999 replications, clustered on state, with the random seeds fixed for reproducibility.

G4. Design-based and sensitivity inference. Randomization inference (Section 5.2) reassigns the eight observed adoption timings to random states in the analytic sample and recomputes the Sun-Abraham average post-adoption effect. Sensitivity to deviations from parallel trends is assessed with the Rambachan and Roth (2023) honest-bounds framework, which asks how large a post-period violation of parallel trends, benchmarked against the pre-period, would have to be before the confidence set for the average post-adoption effect includes zero (Table 3).

Appendix I. Mechanism Atlas

Figures A1–A5 (Exhibits) unpack the governance levers of Figure 2 one at a time; they are conceptual illustrations of the mechanism, not data.

Appendix J. Reported Net Spending and the Statutory-Rebate Ramp

Section 5.7 nets out only the *supplemental* rebate when asking whether capture translates into lower net spending. The natural objection is that the CMS-64 worksheet also reports a directly measured net series (gross drug spending minus *all* rebate offsets: national statutory, state supplemental, and the managed-care counterparts of each) for every state from 2002 through 2024, and that this reported net is the more honest savings outcome. It is not. Used as the dependent variable in the same later-adopter Sun-Abraham specification, the reported all-rebate net produces a large and highly significant decline that is an artifact of federal rebate policy rather than pooling.

Table J1. Net-Spending Outcomes, SSDC Later-Adopter Sun-Abraham

Outcome	Effect	SE
Log gross drug spending (analytic)	−0.041	0.051
Log net of <i>supplemental</i> rebates (Table 6)	−0.060	0.064
Log gross drug spending (reported CMS-64)	−0.039	0.050
Log net of <i>all</i> rebates (reported CMS-64)	−0.378	0.073

Source: Author's calculations from CMS-64 Financial Management Report data; reported gross and net drug spending extracted from the CMS-64 master worksheet.

The reported-gross estimate reproduces the analytic gross series (−0.039 vs. −0.041), confirming the extraction. However, the reported all-rebate net swings to −0.378, an implausible 38-log-point “savings” for a treatment that raises the supplemental rebate share by about 3.4 points.

Three features mark it as contaminated rather than causal.

First, the timing is wrong. The event-time path of the reported net is flat and indistinguishable from zero through the adoption year (event times 0 and 1), then falls discontinuously and keeps growing in magnitude, past −60 log points by event time 2 and beyond −100 at the longest lags, where cohort support is thinnest (Figure 6). A rebate-capture effect on a contemporaneous rebate margin would appear *at* adoption and scale with the roughly three points captured, not arrive with a two-year delay and compound to a doubling of spend. The reported gross path, by contrast, stays flat throughout.

Second, the source of the gap is the statutory rebate, not utilization. The divergence between net and gross lives entirely in the rebate wedge. The implied total rebate share (one minus reported net over reported gross) rises from about 0.19 in 2002 to between 0.58 and 0.73 by 2024, reflecting the post-ACA escalation of inflation-indexed and best-price rebates plus the 2010

extension of rebates to managed-care utilization. This national ramp is common to treated and comparison states and is what year fixed effects exist to absorb.

Third, the ramp is differential across the comparison, which is why the year fixed effects do not absorb it. The mean total rebate share rises by roughly 0.55 for SSDC later adopters (about 0.19 to 0.74) against roughly 0.39 for never-pooled states (about 0.19 to 0.58), a gap of some sixteen points by 2024 - five times the supplemental capture the pool actually delivers. Differences in drug mix and managed-care penetration make the federally set rebate accrue faster in the later-adopting states, and the reported net mechanically books that as a pool “saving.” The supplemental-only net in Table 6 strips this wedge out and is the defensible savings contrast; the reported all-rebate net is shown here only to make clear why it cannot be used.

Appendix K. A Compact Bargaining Model and the Causal Chain

This appendix states, in compact form, the bargaining logic summarized in Section 2.4.

The model. Consider a manufacturer negotiating with a state (or pool) over preferred status for its drug j in some therapeutic class. The value to the manufacturer of winning preferred status in state i is

$$V_{ij} = \Delta q_{ij} \times m_j$$

which is the *steerable volume* Δq_{ij} that preferred status moves toward the drug, times the manufacturer’s margin m_j on that volume. A single state bargaining alone splits V_{ij} with the manufacturer according to their relative bargaining weights; the supplemental rebate is the manufacturer’s payment for avoiding the disagreement outcome (non-preferred placement and lost volume). When states pool, the manufacturer instead bargains against a combined prize

$$V_{pool,j} = \sum_i \Delta q_{ij} \times m_j$$

so the loss from failing to reach a deal (losing preferred placement across the whole bloc) is larger, and the equilibrium rebate rises. The three predictions in Section 2.4 follow directly: the prize aggregates steerable volume, not covered lives, so nominal enrollment without coordinated formulary action adds little; the smallest single-state prizes gain proportionally the most, so smaller states should see larger percentage-point gains; and the prize is largest where close therapeutic substitutes exist, so pooling should help least in protected or sole-source classes.

Pool-type detail. Institutional features sharpen the state-owned-versus-vendor prediction.

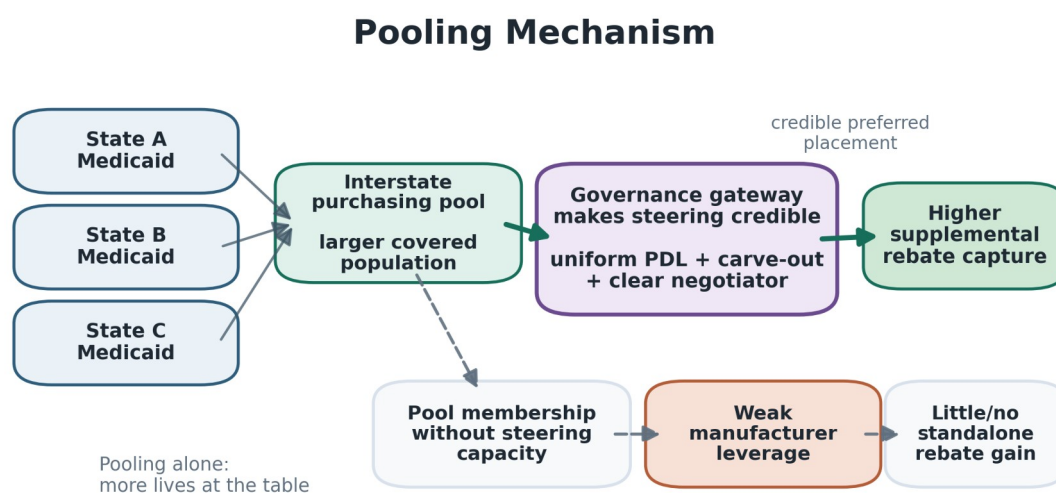
Property-rights logic predicts higher bargaining effort and better adaptation when the party that bears the budget consequence also controls the decisions, as in SSDC. SSDC's state-specific SRAs (Section 2.4) also avoid the "lowest-common-denominator" problem that a single uniform vendor contract can create, and its annual member-state meeting builds a repeated relationship: manufacturers face the same coordinated buyers year after year, which raises the expected cost of lowball offers. The vendor pools' administrator (Prime states that it passes through 100 percent of rebates) faces the objectives described in Section 2.4.

The causal chain. The implied causal chain runs from a state's pre-treatment characteristics, to the decision to enter a pool, through a set of post-treatment governance and contracting changes, to the measured rebate outcome. Pre-treatment confounders sit upstream of pool entry and are valid to adjust for if measured at baseline: baseline rebate capture, Medicaid program size, managed-care penetration, fiscal stress, administrative capacity, drug mix, and the political environment all plausibly drive both who pools and how much rebate a state captures. The treatment node is pool entry itself. Downstream of pool entry sit the *mediators* (a unified PDL, a pharmacy carve-out, expansion of MCO rebate collection, formulary steering, and vendor negotiation), which are the channels through which pooling is supposed to work and which must

not be conditioned on as if they were confounders. The outcomes are the supplemental rebate share (measured here) and, further downstream and not measured in this paper, net drug spending and utilization or access. Adjusting for a baseline characteristic that predates pool entry removes a genuine confounder; conditioning on a post-treatment mediator opens the bad-control problem described in Section 4.2. This classification is the basis for the symmetric control policy in Section 4 and the sensitivity-only treatment of governance controls in Appendix D.

Exhibits

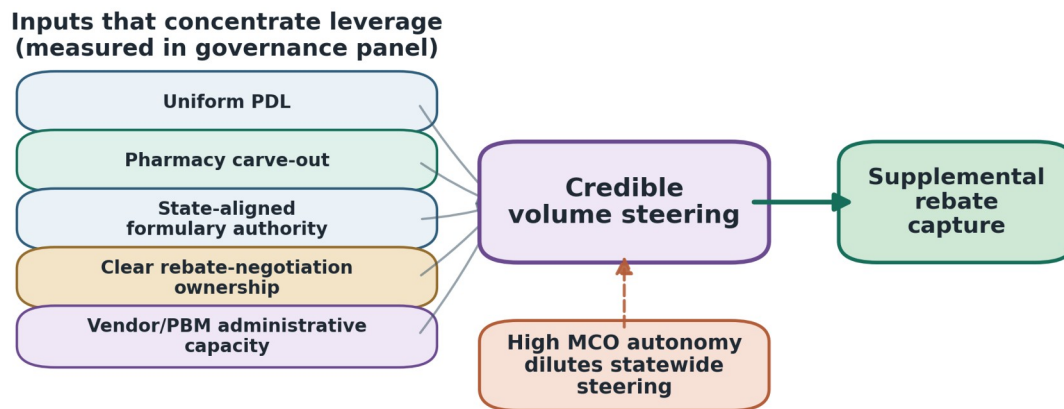
Figure 1. The pooling mechanism



Note: Combining states enlarges the covered population, but higher supplemental rebate capture follows only when the pool can pair that scale with credible volume steering (a unified preferred drug list and a clear negotiator); pooling without steering capacity leaves manufacturer leverage weak.

Figure 2. Governance levers

Governance Levers



Note: The within-state pharmacy-benefit arrangements measured in the governance panel (a uniform preferred drug list, a pharmacy carve-out, state-aligned formulary authority, clear rebate-negotiation ownership, and vendor/PBM capacity) concentrate fragmented formulary control into a single credible volume signal; high managed-care formulary autonomy dilutes it.

Figure 3. Supplemental rebate trends by pool family

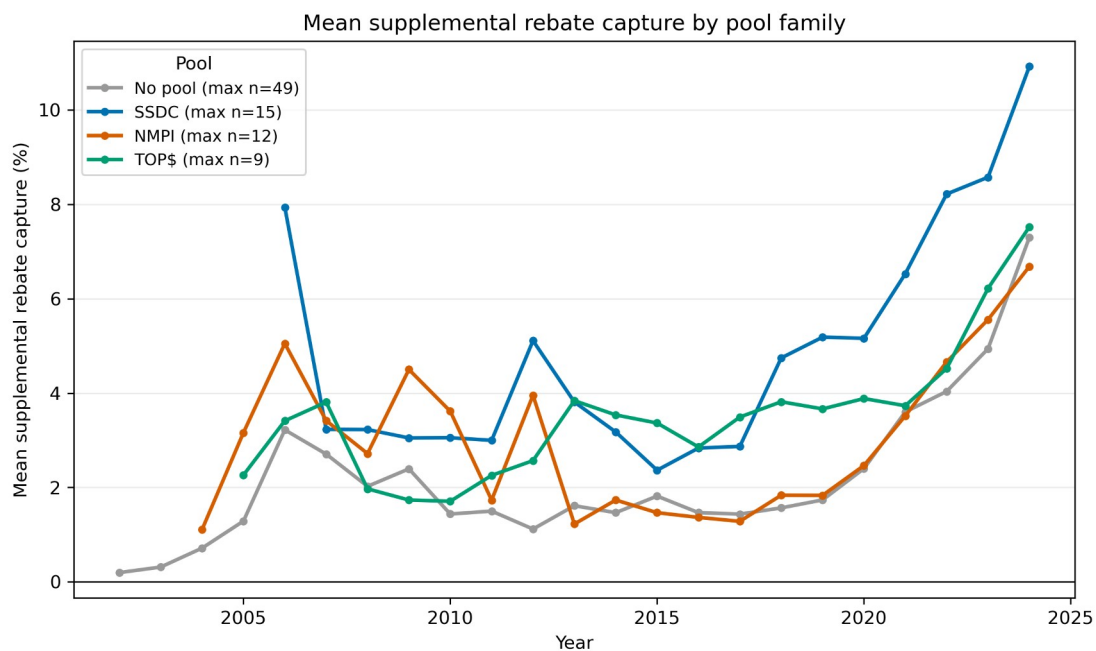


Figure 4. SSDC treatment timing

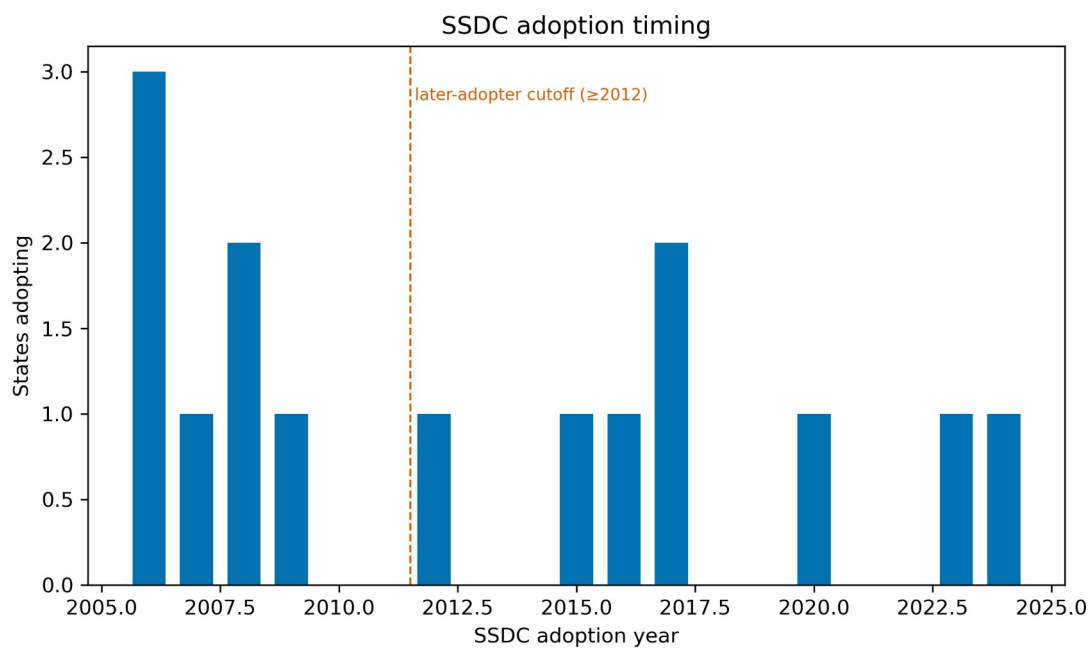


Figure 5. SSDC later-adopter event study (truncated at event time 8)

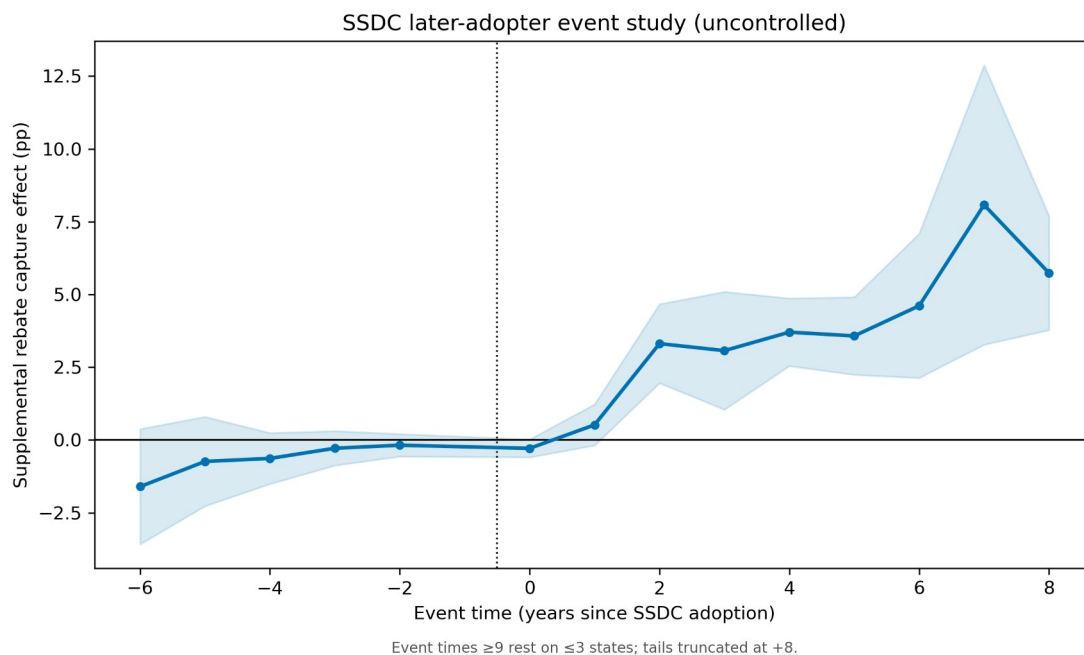
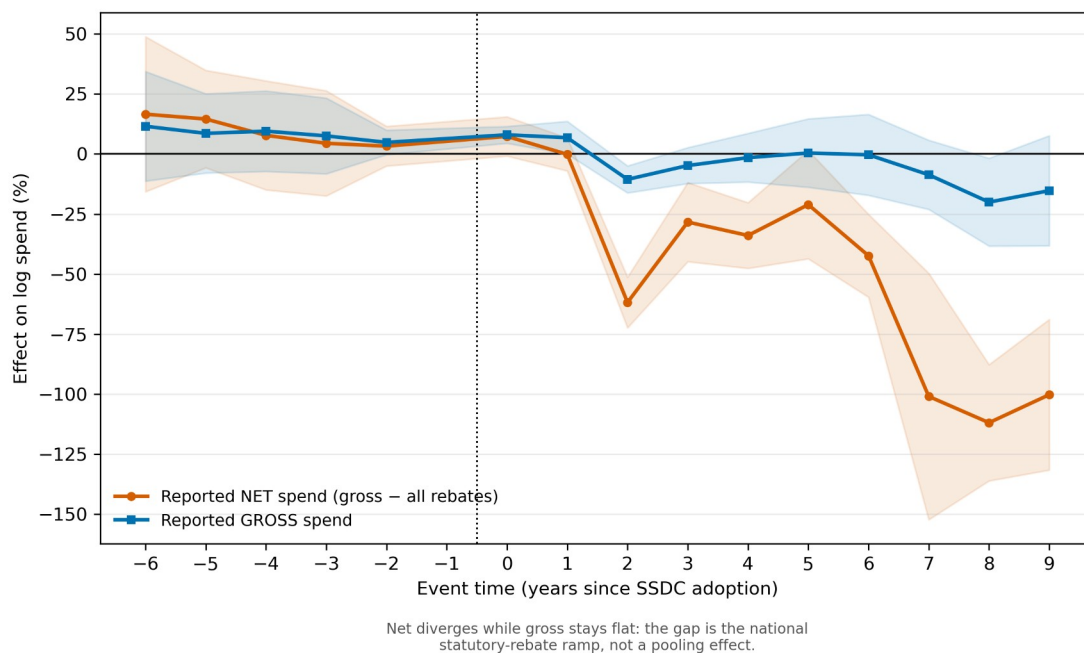
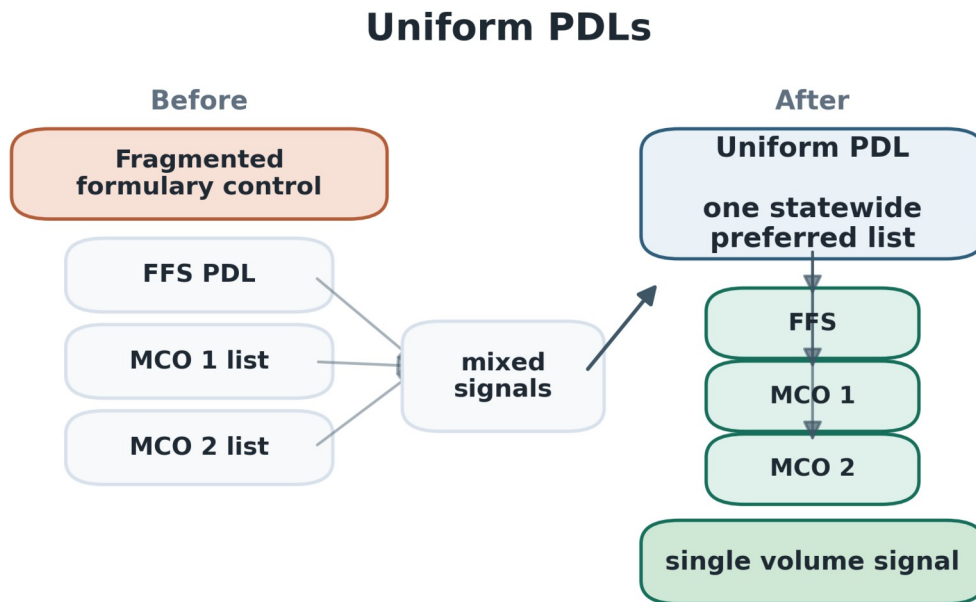


Figure 6. Reported CMS-64 net versus gross drug-spend event-time paths (SSDC later-adopter Sun-Abraham)



Note: The reported all-rebate net diverges sharply and with growing magnitude in the post-period while gross stays flat; the gap is the differential national statutory-rebate ramp, not a pooling effect. The window is truncated to event time -6 to $+9$ for legibility.

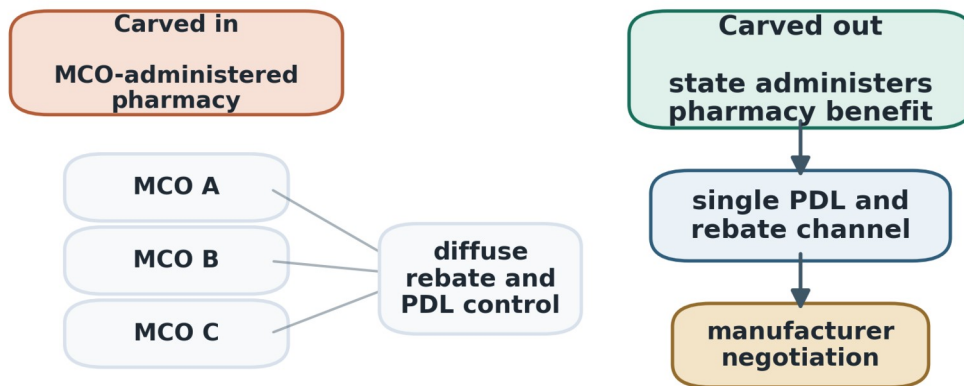
Figure A1. Uniform preferred drug lists



Note: Replacing fragmented fee-for-service and managed-care lists (“mixed signals”) with one statewide preferred list, so a single volume signal stands behind a preferred-status decision.

Figure A2. Pharmacy carve-out

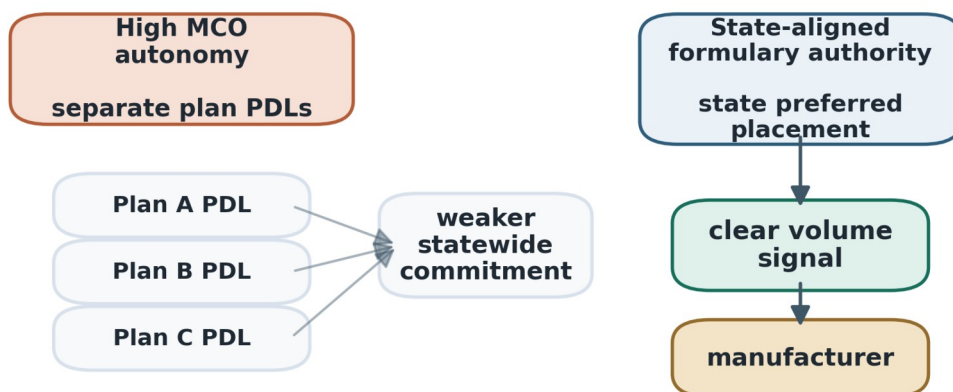
Pharmacy Carve-Outs



Note: Moving the drug benefit from managed care back to fee-for-service centralizes formulary and rebate authority in the state.

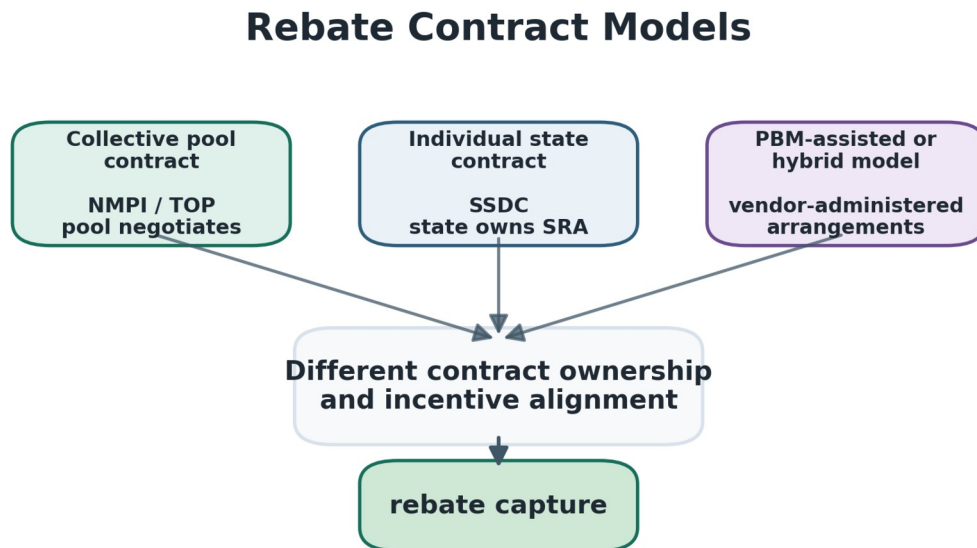
Figure A3. Managed-care formulary autonomy

MCO Formulary Authority



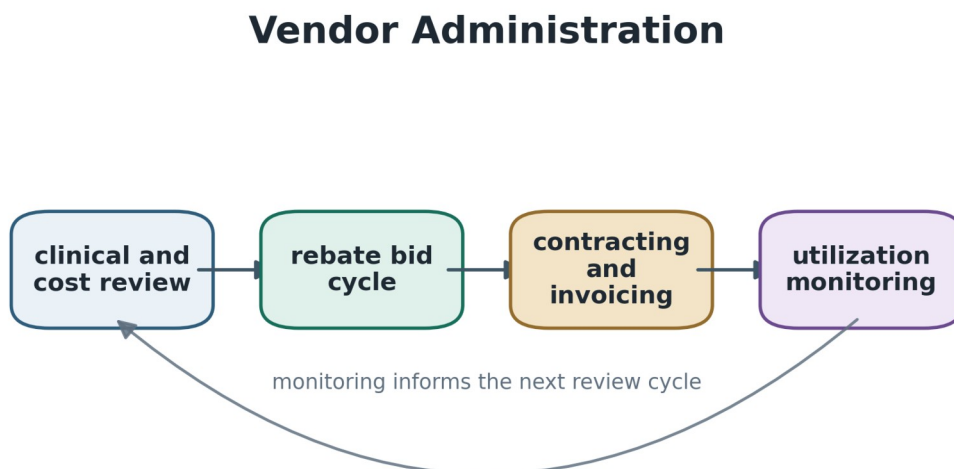
Note: Where managed-care plans retain independent formularies, statewide steering is diluted.

Figure A4. Rebate-negotiation ownership



Note: Who holds the supplemental rebate contract (the state, a pool, or a vendor) shapes how aggressively each incremental rebate dollar is pursued.

Figure A5. Vendor administration



Note: The recurring clinical-review, rebate-bid, contracting, and utilization-monitoring cycle a pool's administrator runs on the state's behalf.